

Lingxin Chen

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Summary

Software engineer with ~3 years building production **C#/.NET** features for **Paratext** (10K+ users), now building **AWS** cloud pipelines, **LLM** evaluation infrastructure, and **PyTorch** graph neural networks at Vanderbilt.

Education

Vanderbilt University | M.S. in Computer Science | GPA: 3.90/4.00

Expected May 2027

Wheaton College | B.S. in Computer Science, cum laude

May 2022

Skills

Languages: Python, C#, Java, TypeScript, JavaScript, SQL

Frameworks / Backend: .NET, FastAPI, Node.js, React, REST APIs, asyncio, OAuth 2.0

AI / Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, Graph Neural Networks, LLMs, vLLM

Cloud / Infrastructure: AWS (Lambda, ECS, EC2, S3, Bedrock, Transcribe, CDK), Docker, Kubernetes, CI/CD

Databases / Tools: PostgreSQL, MySQL, SQLite, Git

Experience

LLM Engineer | *Human Language Analysis Lab, Vanderbilt University* | Nashville, TN

May 2026 – Present

- Built an **LLM generation service** in Python with **8 swappable backends** (Claude, GPT, Gemini, vLLM, llama.cpp) behind one interface, with concurrent, crash-safe resumable runs; generated **15K+ documents**.
- Designed an engine-agnostic SQL layer (**SQLite/MySQL**) feeding the DLATK toolkit and logging every API call.
- Served open-weight LLMs up to **397B parameters** with vLLM and llama.cpp.
- Showed a reported "LLMs lack human-like variance" result was confounded by text length and decoding settings; length-matching raised Llama-4's trait-variance ratio from **0.83 to 1.06** (human = 1.0).

Cloud Engineer | *Vanderbilt Cloud Innovation Lab* | Nashville, TN

Mar 2026 – Sep 2026

- Built an **event-driven AWS pipeline** (S3 → Lambda → ECS) that segments 30-minute archival TV broadcasts into 20+ labeled segments in **~10 min for under \$1/video**.
- Fused shot boundaries (PySceneDetect), speaker-diarized transcripts (Amazon Transcribe), and Claude frame descriptions; merging shots by speaker raised boundary **F1 from 0.04 to 0.44**.
- Defined infrastructure in **AWS CDK** with 0–2 instance autoscaling (\$0 idle), transcript caching, and backoff retries.

Research Assistant | *Sarkar Lab, Vanderbilt University* | Nashville, TN

Nov 2025 – Present

- Developed **SpaPro**, a multimodal **graph neural network (GNN)** in **PyTorch** that predicts protein levels from gene expression and tissue images across **395K cells**, outperforming the best of **8 published methods**.
- Designed the model to train **8–12× faster** than the strongest GNN baseline, enabling **1,000+ logged runs**.
- Rebuilt a leakage-free benchmark of 11 methods; caught a misconfigured baseline that inverted a published ranking.

Full Stack Software Engineer | *SIL Global* | Nashville, TN

Aug 2022 – May 2025

- Built **whitespace & invisible-character support**, the headline feature of **Paratext 9.5**, in C#/.NET across Scripture Reference Settings, Quotation Rules, project creation, and tool menus.
- Resolved **95 issues** across releases 9.3–9.6 (57 bugs, 35 Critical/Urgent), including a data-corruption exception.
- Fixed 25 false-error defects in the text-checking engine and adapted core dialogs for right-to-left UI languages.

Software Engineer Intern | *JAARS* | Waxhaw, NC

Jun 2022 – Aug 2022

- Led a 3-person team building **Support.Bible**, Paratext's community Q&A site for **2,000+ users**, with Google/Facebook sign-in (OAuth 2.0), spam protection, and 11 plugin integrations.
- Migrated the site's database from PostgreSQL to MySQL with zero data loss; deployed on **AWS** (Docker, Kubernetes).

Software Developer Intern | *Yuhuan Rural Commercial Bank* | Taizhou, China

Jun 2021 – Aug 2021

- Built a **React/Node.js** employee portal for 2,000+ users; cut page load time **40%** via caching.

Awards & Publications

- Best Use of AWS Award (\$10,000):** drone and robot disaster-response system on AWS IoT Greengrass and S3.
- 1st Place (\$2,500), AiTinker AI Hackathon:** Domain Sentry, a full-stack app automating DMCA takedowns.
- L. Chen et al.** (first author). *SpaPro: Adaptive RNA-to-Protein Decoding in Spatial Multi-omics Data*. In preparation.